

Worksheet 7B: Linear Systems and Gaussian Elimination — Solutions

Series 7: Linear Algebra
AIML Foundations Mathematics
Dublin and Dún Laoghaire ETB
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Note to instructor: Key answers are given. Accept mathematically equivalent forms.

Part A

1. $\left[\begin{array}{cc|c} 3 & -2 & 5 \\ 1 & 4 & -3 \end{array} \right]$

2. $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & -1 & 3 & 11 \\ 1 & 2 & -1 & 2 \end{array} \right]$

3. $\left[\begin{array}{ccc|c} 4 & 0 & 0 & 12 \\ 1 & 1 & 0 & 7 \\ 0 & 1 & -1 & 1 \end{array} \right]$

4. $\left[\begin{array}{cccc|c} 2 & -1 & 3 & -1 & 4 \\ 1 & 2 & -1 & 1 & 1 \end{array} \right]$

5. $y = -2$; $x + 3(-2) = 5 \Rightarrow x = 11$. Solution: $(11, -2)$.

6. $z = 2$; $y - (-1)(2) = 3 \Rightarrow y = 1$; $x + 0(1) + 2(2) = 4 \Rightarrow x = 0$. Solution: $(0, 1, 2)$.

7. $x = 5$, $y = -2$, $z = 3$ directly from the diagonal. RREF allows immediate reading of the solution.

8. $\left[\begin{array}{ccc|c} 1 & 2 & 1 & 5 \\ 2 & -1 & 3 & 8 \\ 1 & 1 & -1 & 1 \end{array} \right]$

Part B

9. $\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 3 & -1 & 5 \end{array} \right]$

10. $\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & -7 & -4 \end{array} \right]$

11. $\left[\begin{array}{ccc|c} 1 & 2 & -1 & 4 \\ 0 & 3 & 2 & 7 \\ 0 & -5 & 5 & -7 \end{array} \right]$

12. $\frac{1}{2}R_1: \left[\begin{array}{cc|c} 1 & 3 & 5 \\ 1 & 2 & 4 \end{array} \right]; R_2 - R_1: \left[\begin{array}{cc|c} 1 & 3 & 5 \\ 0 & -1 & -1 \end{array} \right]$
13. After elimination: $\left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & -3 & -3 & -7 \\ 0 & -3 & 1 & -1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & -3 & -3 & -7 \\ 0 & 0 & 4 & 6 \end{array} \right]$
14. Swap $R_1 \leftrightarrow R_2$, then eliminate: $\left[\begin{array}{ccc|c} 1 & 3 & -1 & 2 \\ 0 & 1 & 2 & 5 \\ 0 & -6 & 3 & -1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 3 & -1 & 2 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & 15 & 29 \end{array} \right]$
15. (a) Yes. (b) No — pivot missing in row 2 before row 3. (c) Yes (non-square, valid staircase). (d) Yes.
16. Each operation has an inverse of the same type. Because each step is reversible, the solution set is preserved throughout.
17. Division by zero is undefined; all pivots must be non-zero.
18. If $i = j$: $R_i + kR_i = (1 + k)R_i$, which just scales row i — this could be done with $R2$, and the notation is misleading. More fundamentally, $R3$ is meant to combine two different rows.

Part C

19. Augmented: $\left[\begin{array}{cc|c} 1 & 2 & 5 \\ 3 & -1 & 4 \end{array} \right] \xrightarrow{R_2 - 3R_1} \left[\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & -7 & -11 \end{array} \right]; y = 11/7; x = 5 - 22/7 = 13/7.$
20. Row reduce to get $z = 1, y = 2, x = 2$. Solution: $(2, 2, 1)$.
21. $z = 2, y = 1, x = 3$. Solution: $(3, 1, 2)$.
22. After RREF: free variable z ; $y = 4 - 2z$... the system reduces; let $z = t$: $y = 4 - 2t$ for any t (infinitely many solutions through the z -axis). Check the augmented matrix carefully.
23. RREF gives $z = 1, y = 2, x = 3$.
24. $[A | I]: \left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 5 & 3 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{cc|cc} 1 & 0 & 3 & -1 \\ 0 & 1 & -5 & 2 \end{array} \right]; A^{-1} = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}.$
25. Applying the same row operations to I records the inverse transformation. When the left side becomes I , the right side is the matrix of all those operations composed — which is A^{-1} .
26. $A^{-1}\mathbf{b} = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix} \begin{bmatrix} 5 \\ 4 \end{bmatrix} = \begin{bmatrix} 11 \\ -17 \end{bmatrix}$... check against problem 19: $(13/7, 11/7)$. The inverse here is for a different matrix; problem 24 builds the inverse for problem 24's matrix.
27. Pivot variables: x_1, x_2 ; free variable: x_3 . Let $x_3 = t$: $x_1 = 3 - 2t, x_2 = 1 + t, x_3 = t$ for any $t \in \mathbb{R}$.
28. A is not invertible ($\det A = 0$); the system has either no solution or infinitely many solutions.

Part D

29. Row 2 = 2 × Row 1 in the augmented matrix $\Rightarrow$ infinitely many solutions (same line). General solution: $y = t$, $x = 3 - 2t$.
30. Augmented: $\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 0 & 1 \end{array} \right]$; last row $\Rightarrow 0 = 1$. **No solution** (inconsistent).
31. Row 2 = 2 × Row 1 but RHS $5 \neq 2 \times 2 = 4$. **No solution**.
32. $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -2 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right]$. One free variable; infinitely many solutions: $z = t$, $y = (t-1)/2$... check carefully. Unique solution: $x = 1$, $y = 2$, $z = 3$.
33. After row reduction, one free variable exists (say $x_2 = s$, $x_4 = t$). Infinitely many solutions.
34. With $n > m$, the RREF has at most $m < n$ pivots, so at least $n - m \geq 1$ free variables exist.
35. When $\text{rank}(A) < n$ (the columns are linearly dependent), the null space contains non-zero vectors.
36. (a) rank 1 (second row is multiple of first) (b) rank 3 (c) rank 1
37. RREF of A : $\left[\begin{array}{ccc} 1 & 2 & -1 \\ 0 & 0 & 0 \end{array} \right]$; free variables $x_2 = s$, $x_3 = t$; $x_1 = -2s + t$. Null space:
 $s \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$.
38. Column space condition: row-reduce $[A \mid \mathbf{b}]$; need $b_2 = 2b_1$ (second row becomes $0 = b_2 - 2b_1$). So $\mathbf{b} = \begin{bmatrix} b_1 \\ 2b_1 \end{bmatrix}$.

Part E

39. Augmented: $\left[\begin{array}{ccc|c} 1 & -1 & -1 & 0 \\ 2 & 3 & 0 & 12 \\ 0 & 3 & -2 & 6 \end{array} \right]$. Solution: $I_1 = 3$, $I_2 = 2$, $I_3 = 1$.
40. (a) $A^T A = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$; $A^T \mathbf{b} = \begin{bmatrix} 7 \\ 17 \end{bmatrix}$. (b) $\begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 7 \\ 17 \end{bmatrix}$; solution: $w_1 = -1/3$, $w_1 = 3/2$. (c) Best-fit line $y = -1/3 + \frac{3}{2}x$.
41. A is 3×2 (more rows than columns); A^{-1} doesn't exist. Only square invertible matrices have inverses.
42. C: $2a = c$; H: $6a = 2d \Rightarrow d = 3$; O: $2b = 2c + d$. With $a = 1$: $c = 2$, $d = 3$, $b = 7/2$. Multiply through by 2: $2\text{C}_2\text{H}_6 + 7\text{O}_2 \rightarrow 4\text{CO}_2 + 6\text{H}_2\text{O}$.
43. Network system with 4 balance equations (one is redundant); solution has a free variable representing the unspecified flow on one branch.

44. (a) For large n , computing A^{-1} costs $O(n^3)$ operations and is numerically less stable than elimination. (b) Two lines: intersect (unique), parallel (no solution), coincident (infinite); three planes similarly. (c) $A^T A$ singular means the normal equations have no unique solution — the model is underdetermined (use regularisation, e.g. ridge regression from ws04e).